Study Notes: Overview of Java

Introduction to Java

Java is a high-level, class-based, **object-oriented programming language** designed to have as few implementation dependencies as possible. It was developed by **James Gosling** at Sun Microsystems and released in **1995**.

Core Philosophy

- **WORA (Write Once, Run Anywhere):** Java code is compiled into an intermediate form called **Bytecode**, which can run on any device equipped with a Java Virtual Machine (JVM).
- **Platform Independence:** Unlike languages like C++, Java does not compile into machine-specific code but into platform-neutral bytecode.

Key Features of Java

To excel in exams, remember these "buzzwords" that define Java's architecture:

- **Simple:** Removes complex features like explicit pointers and operator overloading.
- **Object-Oriented:** Everything in Java is an **Object**, which allows for modular and extensible code.
- **Robust:** Emphasizes early checking for possible errors, with a strong focus on **Memory Management** and **Exception Handling**.
- **Secure:** Java runs inside a "sandbox" (the JVM), preventing unauthorized access to system resources.
- **Multi-threaded:** Allows programs to perform many tasks simultaneously by defining multiple threads of execution.

Java Architecture: JDK, JRE, and JVM

Understanding the distinction between these three components is critical for technical exams.